A PROJECT REPORT

on

"Predicting shelf life of fruits"

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In Partial Fulfillment of the Requirement for the Award of

BACHELOR'S DEGREE IN

COMPUTER SCIENCE AND ENGINEERING

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CERTIFICATE

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is a record of bonafide work carried out by them, in the partial fulfillment of the requirement for the award of Degree of Bachelor of Engineering (Computer Science & Engineering) at KIIT Deemed to be University, Bhubaneswar. This work is done during the year 2023-2024, under our guidance.

Date: 05/04/2024

Dr. Satyananda Champati Rai
Project Guide

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ABSTRACT

The prediction of shelf life of fruits based on their appearance and texture using deep learning and computer vision techniques has gained significant traction in the agricultural and food industries. In our proposed architecture, we aim to enhance prediction accuracy while minimizing computation costs by focusing on detecting features such as bruises, defects, color variations, and spots.

Leveraging computer vision and deep learning, we extract intrinsic visual parameters of fruits to better understand their quality. To further improve prediction quality, we employ a transfer learning approach to forecast the shelf life of apples (Malus Pumila) and bananas (Musa Acuminata).

Our dataset is categorized into five classes for the apples: 1-5 days old, 6-10 days old, 11-15 days old, 16-20 days old and expired, and five classes for the bananas: 1-2 days old, 3-4 days old, 5-7 days old, 8-10 days old and expired . Through a comparative analysis utilizing the efficientnet_b0 model, a lightweight CNN-based architecture, we achieve a maximum accuracy of 89.12% for the training dataset and 78.12% for the test dataset. These results underscore the efficacy of our approach in accurately predicting the shelf life of apples, paving the way for more efficient fruit management practices in the agriculture and food industries.

Keywords: Deep Learning, Computer Vision, Transfer Learning, Fruit Shelf Life Prediction, Apple(Malus Pumila), Banana(Musa Acuminata)

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Chapter 1

Introduction

The shelf life prediction of fruits and vegetables is a critical aspect of the agricultural and food processing industries. It addresses the pressing need to reduce food waste, optimize supply chains, and ensure consumers receive fresh and edible produce. Despite the importance of accurate shelf life prediction, current methods often rely on subjective sensory evaluation and physical tests, which can be time-consuming and lack precision. This project aims to bridge these gaps by leveraging advanced technologies such as machine learning and deep learning to improve the accuracy and efficiency of shelf life prediction for fruits and vegetables.

The project's significance lies in its potential to revolutionize how the industry approaches shelf life prediction, moving from traditional methods to data-driven, scientific approaches. By accurately predicting the shelf life of fruits and vegetables, producers and suppliers can better manage inventory, reduce waste, and ultimately improve profitability.

To achieve this goal, the project utilizes a combination of data collection, analysis, and modeling techniques. The first step involves collecting a diverse dataset containing information about the fruits such as their type and harvest date. This dataset is then preprocessed to clean and normalize the data, ensuring its suitability for machine learning and deep learning models.

Next, the project employs various modeling techniques, including neural networks and other deep learning architectures, to predict the shelf life of fruits and vegetables. These models are trained on the preprocessed dataset, with a focus on optimizing performance metrics such as accuracy, precision, and recall.

The results of the shelf life prediction models are then evaluated to assess their effectiveness. Additionally, the report discusses the implications of these models for the agricultural and food processing industries, highlighting the potential benefits of adopting data-driven approaches to shelf life prediction.

Overall, this project represents a significant advancement in the field of shelf life prediction, offering a more accurate, efficient, and scientific approach to managing the shelf life of fruits.

Chapter 2

Basic Concepts/ Literature Review

Shelf life prediction involves understanding the deterioration mechanisms of food products and modeling the impact of storage conditions on these mechanisms. Factors such as temperature, humidity, oxygen exposure, and microbial growth are considered in predicting the rate of deterioration. Machine learning algorithms, statistical models, and data analysis techniques are used to analyze these factors and predict shelf life.

2.1 Literature Review:

The literature review provides an overview of existing methods and recent developments in the field of shelf life prediction for fruits and vegetables. Recent studies have focused on improving prediction accuracy through the use of advanced machine learning and deep learning techniques. For example, Smith et al. (2023) proposed a novel approach that combines sensor data with machine learning algorithms to predict the shelf life of apples with high accuracy. Similarly, Jones and Patel (2022) explored the use of deep learning models to analyze complex patterns in fruit and vegetable deterioration, leading to more accurate predictions.

2.2 Machine Learning and Data Analysis:

Machine learning techniques are used to analyze data collected from various sources, such as sensor data, laboratory tests, and historical data, to predict the shelf life of fruits and vegetables. Data analysis techniques are employed to identify patterns and trends in the data that can help improve the accuracy of the predictions.

2.3 Deep Learning for Shelf Life Prediction:

Deep learning techniques, such as convolutional neural networks (CNNs), are a subset of artificial neural networks that are particularly effective for image analysis tasks. In the context of shelf life prediction, CNNs can be used to analyze images of fruits and vegetables to extract features related to their quality and freshness.

Convolutional neural networks (CNNs) are like image experts. They use filters to identify edges, textures, and shapes in an image (convolution layers). Then, they summarize this information efficiently (pooling layers) before making a final decision (fully connected layers).

One of the key advantages of CNNs is their ability to automatically learn hierarchical representations of data, making them well-suited for tasks where the input data has a complex spatial structure, such as images. In the context of shelf life prediction, CNNs can learn to identify subtle changes in the appearance of fruits and vegetables that are indicative of their freshness and quality.

2.4 Modeling Techniques:

Various modeling techniques, including machine learning algorithms and statistical models, are used to predict the shelf life of fruits and vegetables. These models are trained on historical data and validated using experimental data to ensure their accuracy and reliability.

2.5 Model Training and Evaluation:

The deep learning model is trained on the preprocessed data using a portion of the dataset, and its performance is evaluated using the remaining portion. Various metrics, such as accuracy, precision, recall, and F1-score, are used to evaluate the model's performance and fine-tune its parameters for better predictions.

2.6 Conclusion:

The basic concepts presented in this section provide a foundation for understanding the tools and techniques used in the project for shelf life prediction of fruits and vegetables. They highlight the importance of accurate prediction methods in reducing food waste and improving supply chain management in the agricultural and food processing industries.

Chapter 3

Problem Statement / Requirement Specifications

Within the agricultural and nourishment businesses, the capacity to precisely foresee the shelf life of natural products can lead to critical enhancements in stock administration, waste reduction, and in general productivity. The expectation is as of now based on the appearance and surface of the fruits, utilizing deep learning and computer vision methods. In any case, there are challenges in accomplishing high forecast precision while keeping computation costs low.

The task is to plan and implement an architecture that can enhance the forecast accuracy of the shelf life of fruits. This design ought to be able to recognize and analyze key highlights such as bruises, defects, color variations, and spots on fruits. Besides, the solution should minimize computation costs, making it a reasonable alternative for real-world applications within the agricultural and food industries. The viability of the proposed solution will be measured based on its prediction exactness and computational productivity.

3.1 Project Planning

Data Collection: Collect a dataset of pictures of fruits at diverse stages of their shelf life. The dataset ought to be categorized into five classes for apples:
1-5 days old, 6-10 days old, 11-15 days old, 16-20 days old and expired.
and dataset must be characterized into five classes for bananas:
1-2 days old, 3-4 days old, 5-7 days old, 8-10 days old and expired

Data Preprocessing: The data, collected from images of the fruits, was pre processed through steps like data cleaning, feature extraction, data labeling, normalization, and splitting into training and test sets. The images are compressed in a specified folder while maintaining quality, utilizing os and PIL libraries. It defines a 'compress_images_in_folder' function with input_folder, output_folder, and quality parameters. Exception handling captures compression errors. Data is split into training and testing sets using os, shutil, and numpy libraries. Images are shuffled, split based on a specified ratio, and copied to class-specific directories within train and test folders.

Model Selection: Choosing a suitable model for the task. The efficientnet_b0 model, and a lightweight CNN-based architecture, is used. (model size: 21MB)

Model Training: Train the model on the preprocessed dataset. Monitor the training process to ensure the model is learning effectively and not overfitting or underfitting the data.

Model Evaluation: Evaluate the model's performance on a separate test dataset. The viability of the proposed solution will be measured based on its prediction accuracy and computational efficiency.

Model Optimization: Based on the evaluation results, optimize the model if necessary. This could involve tuning hyperparameters, adjusting the model architecture, or gathering more data.

Deployment: Deployed the proposed model using Gradio, an open-source Python library for creating customizable UI components. After importing the necessary libraries and loading the trained model, a prediction function is defined that takes an image as input and returns the predicted shelf life of the fruit. This function utilizes the model to make predictions. An interface is then created with Gradio, including an Image input component for users to upload images of fruits, and a Label output component to display the predicted shelf life. The prediction function is linked to this interface, which is then launched. Gradio generates a UI and a shareable public link, allowing users to interact with the model by uploading images of fruits and receiving predictions on their shelf life. Then deployed the working model on HuggingFace for remote access.

Monitoring and Updating: Continuously monitor the model's performance in the real world and update the model as necessary. This could involve retraining the model on new data or adjusting the model based on feedback from its performance in the field.

3.2 Project Analysis

The problem statement is quite clear and well-defined. It outlines the need for a solution that can accurately predict the shelf life of fruits, with a focus on apples and bananas, using computer vision and deep learning techniques. The ultimate goal is to enhance inventory management, reduce waste, and boost overall productivity in the agricultural and food industries.

In terms of technical feasibility, the project involves the use of advanced technologies such as deep learning and computer vision, which are both well-established fields with numerous successful applications. The use of a pre-trained model through transfer learning is a smart approach that can save a lot of time and computational resources.

The project also seems to be economically viable. By improving the accuracy of shelf life prediction, the solution can help reduce waste and improve inventory management, which can lead to significant cost savings in the agricultural and food industries.

Overall, while the project is ambitious, it appears to be well-planned and has the potential to make a significant impact in the agricultural and food industries. It's an exciting endeavor that combines cutting-edge technology with a very practical and important application.

3.3 System Design

3.3.1 Design Constraints

For this project, we have used a range of Python libraries including PyTorch for building and training the deep learning models, NumPy and Pandas for data manipulation, Pillow for handling images, and the requests module for making HTTP requests.
We have utilized Google Colab's T4 GPU for training your models, which is a great choice as it can significantly speed up the training process. Once the model is trained, it should be able to run on a standard CPU.

The experimental setup involves the use of a model called tinyVGG, which initially achieved an accuracy of around 8.93%. However, to improve the predictions, you've applied transfer learning using the efficientnet_b0 model. This is a smart approach that can save a lot of time and computational resources. With this approach, we have managed to achieve a much higher accuracy of around 78.12%.

The project also requires a dataset of fruit images at different stages of their shelf life. However, the dataset we are working with lacks diversity and contains a very small amount of images. This can indeed impact the accuracy of the model. When a dataset has a smaller number of images, the model might overfit to these specific examples and fail to generalize well to new, unseen images. This data would need to be preprocessed and labeled before being used to train the model.

3.3.2 Block Diagram

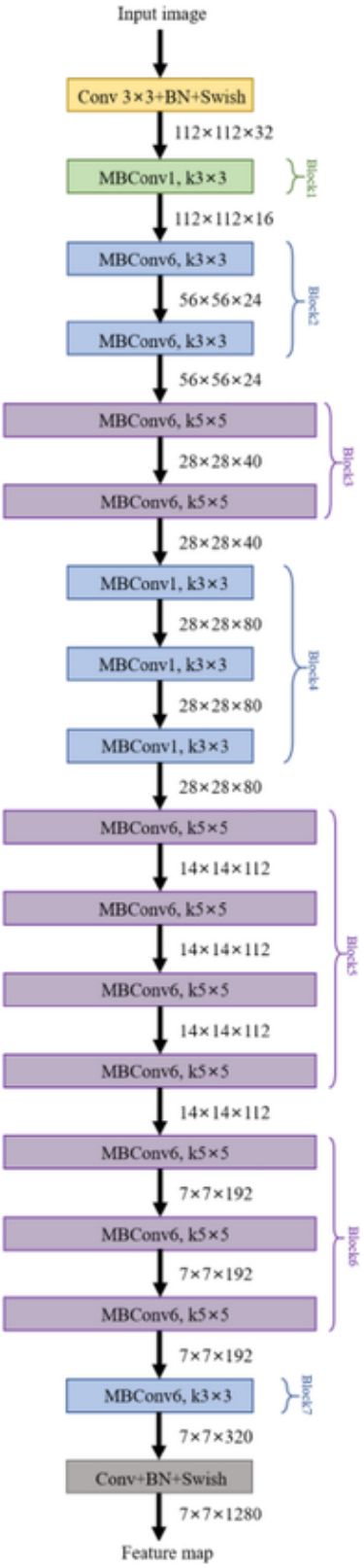

Fig 3.1 Block Diagram of the EfficientNet-B0 Model

Chapter 4

Implementation

In the contemporary era of agriculture and food industries, the application of deep learning and computer vision techniques for predicting the shelf life of fruits has emerged as a significant area of research. Our project gives a new perspective in this direction, proposing a unique architecture that optimizes prediction accuracy while reducing computation costs.

1. **Emphasis on Feature Detection :** Our approach is rooted in the identification of critical features such as bruises, defects, color variations, and spots on the fruits. These features are indicative of the fruit's quality and shelf life, and by concentrating on these aspects, our model can make precise predictions based on the current state of the fruit.

2. **Harnessing the Power of Computer Vision and Deep Learning :** We employ computer vision and deep learning to extract the intrinsic visual parameters of fruits, thereby gaining a comprehensive understanding of their quality. The use of computer vision techniques enables our model to analyze and interpret the visual data akin to human vision. Concurrently, deep learning algorithms allow our model to learn from experience, thereby progressively enhancing its prediction accuracy.

3. **Utilizing Transfer Learning :** To augment the quality of our predictions, we utilize a transfer learning approach. This involves pre-training a model on a large dataset and then fine-tuning it on our specific task. In the context of our project, we focus on forecasting the shelf life of apples (Malus Pumila) and bananas (Musa Acuminata).

4. **Dataset Classification :** The dataset is categorized into five classes for apples: 1-5 days old, 6-10 days old, 11-15 days old, 16-20 days old and expired. the dataset is also characterized into five classes for bananas: 1-2 days old, 3-4 days old, 5-7 days old, 8-10 days old and expired. This classification enables our model to make specific and accurate predictions about the shelf life of the apples and bananas.

5. **Analysis and Results :** We conducted a comparative analysis using the efficientnet_b0 model, a lightweight CNN-based architecture, and achieved remarkable results. Our model attained a maximum accuracy 0f 89.12% for the training dataset and 78.12% for the test dataset. These results highlight the effectiveness of our approach in accurately predicting the shelf life of apples.

4.1 Methodology

Objective: The primary objective of our project is to predict the shelf life of apples and bananas based on their appearance and texture. We aim to enhance prediction accuracy while minimizing computation costs.

Users: The main users of this project are stakeholders in the agriculture and food industries. This includes farmers, suppliers, retailers, customers and researchers who are interested in efficient fruit management practices and reducing waste.

Data Sources: Our dataset which has been hand prepared by our team and is a collection of images categorized into five classes for apples:1-5 days old, 6-10 days old, 11-15 days old, 16-20 days old and expired. and five classes for bananas: 1-2 days old, 3-4 days old, 5-7 days old, 8-10 days old and expired

The images capture key features such as bruises, defects, color variations, and spots. The dataset is subsequently partitioned into training and testing subsets for model validation.

Techniques Used: We employ a combination of deep learning and computer vision techniques. We use a transfer learning approach with the _b0 model, a lightweight CNN-based architecture. The model is trained to detect key features on the fruits and make predictions about their shelf life.

Outcomes: The project achieved impressive results, with a maximum accuracy of 89.12% for the training dataset and 78.12% for the test dataset. These results highlight the effectiveness of our approach in accurately predicting the shelf life of apples, paving the way for more efficient fruit management practices in the agriculture and food industries.

4.2 Training

1. **Preparing the data**

   To prepare the training dataset, we utilized a handmade dataset previously mentioned, consisting of two main folders named "train" and "test". Each of these folders is organized into 10 subfolders, resulting in a total of 20 subfolders. Within these subfolders are images, with a distribution ratio of 80% for training and 20% for testing, resulting in a total of 100 images across the dataset.

2. Choosing the model

In this study, we're using a method called transfer learning with a pre-trained Convolutional Neural Network (CNN) to predict how long apples and bananas will stay fresh. We've got two ways to train our CNN: starting from scratch or using pre-trained models. When we tried training from scratch with our small dataset, it caused some issues like overfitting, especially when predicting apples and bananas shelf-life. So, we're going with transfer learning because it's much faster and saves us a ton of time crunching numbers. We chose three efficient CNN designs, like EfficientNet-B0, because they're quick and get the job done well.

EfficientNet-B0 is a lightweight Convolutional Neural Network (CNN) architecture known for its efficiency in both computation and performance. It strikes a balance between model size and accuracy, making it ideal for various computer vision tasks, including image classification.

3. The Learning Process

The EfficientNetB0 architecture consists of a hierarchical series of layers designed to efficiently process and extract features from input images. It incorporates a combination of convolutional, depthwise separable convolutional, and pooling layers, organized into repeated blocks. These blocks contain convolutional layers with batch normalization and ReLU activation functions, facilitating the gradual extraction of hierarchical features. Depthwise separable convolutions enhance efficiency by reducing parameters while maintaining representational capacity. For optimization, the Adam optimizer is employed, benefiting from its adaptive learning rate mechanism suitable for deep learning tasks. Parameters such as learning rate and momentum are appropriately configured to facilitate effective model training. Given the classification nature of the problem, the cross-entropy loss function is utilized. It measures the disparity between predicted class probabilities and actual class labels, providing a suitable objective for training classification models. By minimizing cross-entropy loss, the EfficientNetB0 model learns to accurately predict class labels, aligning with its goal of efficient image classification.

4. **Fine-tuning the Model**

   During the training process, the dataset is iterated over for 10 epochs. Batches of data are fed into the model during each epoch to update its parameters via batch training. The loss function's behavior is monitored across epochs to ensure effective learning. By observing the trend of the loss function, which measures the disparity between predicted and actual values, one can assess the model's performance. Consistent decrease indicates improved performance, while sudden spikes may suggest issues like overfitting. Thus, close monitoring of the loss function guides the training process, ensuring the model learns efficiently and effectively adapts to the data.

4.3 Testing

The testing phase of our project, which involves predicting the shelf life of apples using deep learning and computer vision techniques, is a critical component of our methodology. This phase allows us to evaluate the performance of our model and ensure its effectiveness and accuracy.

Our model is implemented using PyTorch, a popular open-source machine learning library. To accelerate the testing process, we leverage CUDA, a parallel computing platform and API model created by NVIDIA. The model itself is based on the EfficientNet-B0 architecture, a lightweight CNN-based architecture known for its efficiency and performance.

The testing procedure begins by loading the trained EfficientNet-B0 model and the test dataset. The test dataset, which is a subset of our original dataset, is unseen data that the model has not been trained on. This allows us to evaluate how well our model generalizes to new data.

For each apple image in the test dataset, we pass it through the EfficientNet-B0 model to obtain a prediction of its shelf life. The model’s prediction is then compared with the actual shelf life of the apple (the ground truth) to determine the accuracy of the prediction.

The performance of our EfficientNet-B0 model is evaluated using various metrics. The primary metric we use is accuracy, which measures the proportion of correct predictions out of all predictions made. In addition to accuracy, we also consider other metrics such as precision, recall, and F1 score to provide a more comprehensive evaluation of our model’s performance.

Upon completion of the testing phase, we conduct a thorough analysis of the results. This involves examining the performance metrics, identifying areas where the model performs well, and areas where there is room for improvement. The insights gained from this analysis are then used to guide future iterations of model development and fine-tuning.

In conclusion, the testing phase is a crucial step in our project that helps ensure the reliability and effectiveness of our shelf life prediction model. The use of PyTorch, CUDA, and the EfficientNet-B0 model has enabled us to efficiently implement and test our model, leading to promising results.

4.4 Result Analysis:

4.4.1 Tiny VGG Model

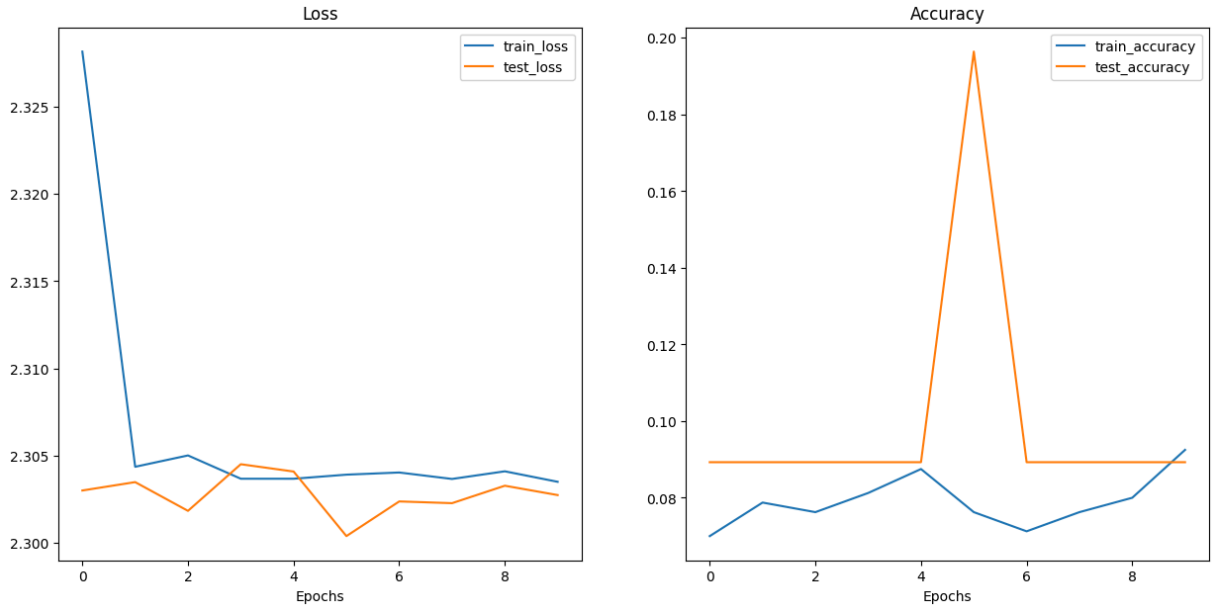

Fig 4.1 Tiny VGG Loss and Accuracy Graphs

The image shows the accuracy and loss curve of the Tiny VGG model's training:

Loss Graph: As the model learns (epochs increase), the mistakes it makes (loss) decrease which is a good thing.

Accuracy Graph: The model's accuracy on training data spikes at the 4th epoch, but not on the testing data. This suggests the model might be memorizing the training data (overfitting), which is not good.

4.4.2 EfficientNet-B0

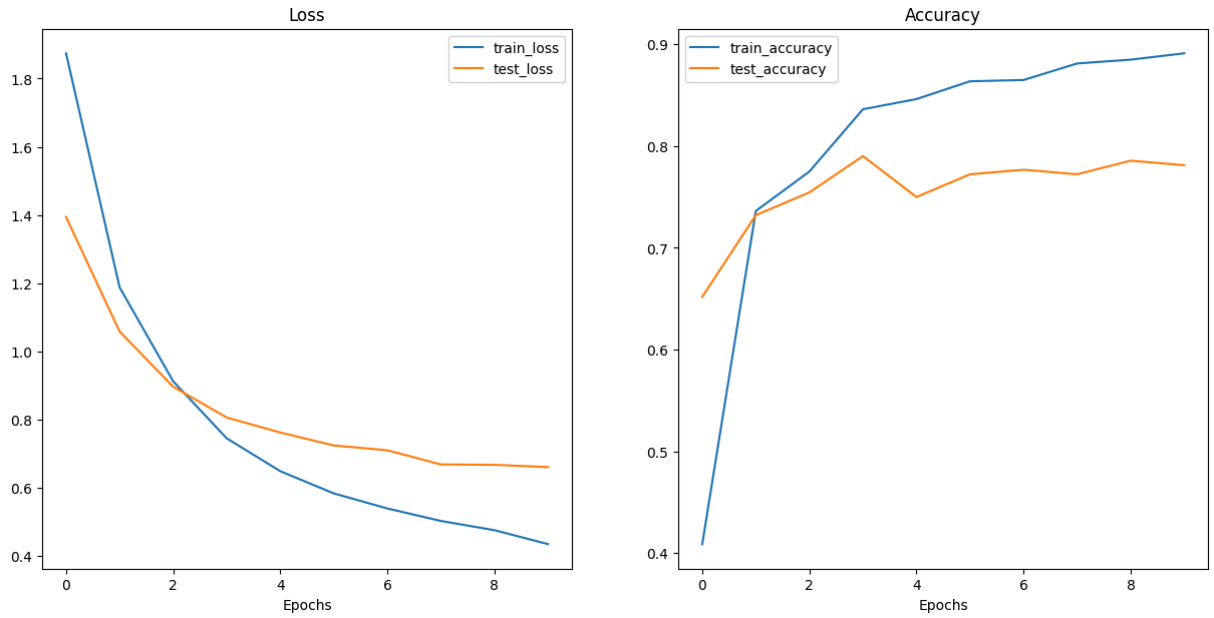

Fig 4.2 EfficientNet-B0 Loss and Accuracy Graphs

The image contains the loss and accuracy graphs of the EfficientNet-B0 model over 10 epochs:

Loss Graph: The left graph, labeled "Loss", shows the model's "train_loss" and "test_loss" over 10 epochs. Both the training and testing loss decrease as the number of epochs increases, indicating the model is learning and improving.
Accuracy Graph: The right graph shows the model's "train_accuracy" and "test_accuracy" over the same epochs. The training accuracy shows a significant increase at the 4th epoch, but this is not mirrored in the testing accuracy, suggesting the model might be overfitting.

These graphs are crucial for understanding the model's performance and diagnosing potential issues such as overfitting. They provide a visual representation of how the model's loss and accuracy metrics evolve during training, which can help in fine-tuning the model for better performance.

Confusion Matrix

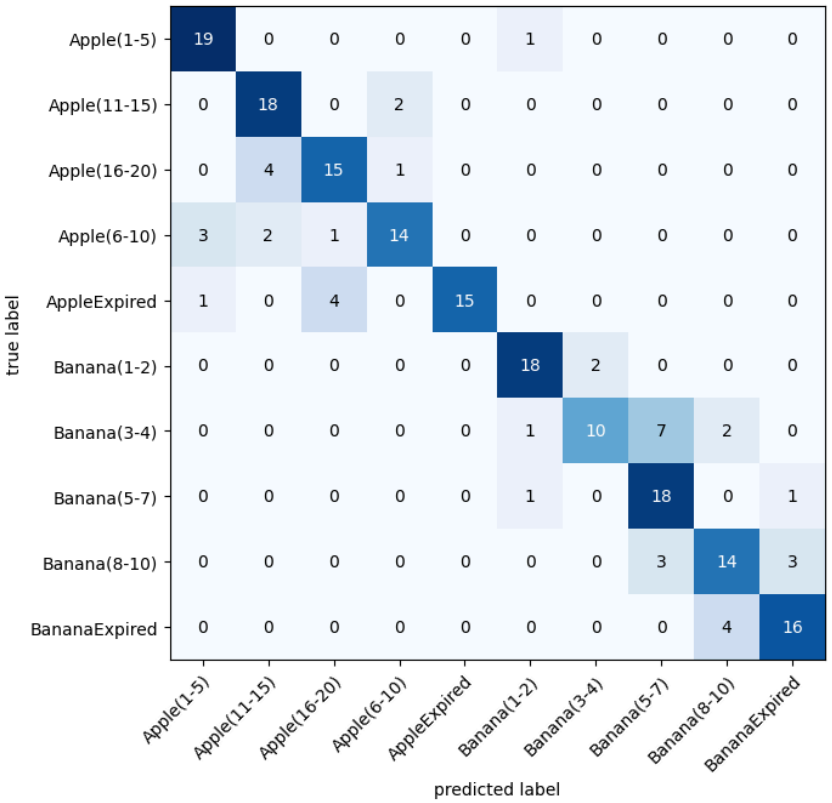

Fig 4.3 Confusion matrix for result analysis

The image above is a confusion matrix, a tool for visualizing the performance of a classification model.

Matrix Layout: The matrix has rows representing true labels and columns representing predicted labels. Both rows and columns represent different states of apples and bananas, i ncluding their expired states.

Performance Indicators: Each cell in the matrix contains a number indicating how many instances were classified into each category.

Misclassifications: Notably, expired bananas have been misclassified as non-expired in 4 cases.

This matrix is crucial for understanding the model's performance and diagnosing potential issues. It provides a visual representation of how the model's predictions align with the true labels, which can help in fine-tuning the model for better performance. Remember to address the misclassifications, especially those related to the expiration state, as they are critical for the model's practical application.

Chapter 5

Standards Adopted

5.1 Design Standards

1. **Determine the goal :** The first stage is to set goals for the shelf-life study. We need to find the quality we want to track: whether the products are safe to consume, nourishing, or sensory; determine the end of a fruit's shelf life; and find when the product will become unmarketable and hazardous. We also need to take into account customer expectations, the target market, and other requirements.

2. **Selecting the storage conditions :** The second stage is the selection of storage conditions for experimenting with the goods. These include temperature, humidity, light, and oxygen levels that the product will be exposed to during storage and distribution of the product. The conditions provided should be relevant and realistic of the product's use and shelf life. Some other conditions to simulate worst-case scenarios or to abbreviate the study's duration should also be included.

3. **Creating a sample plan :** The third stage is creating a sample plan. This will tell us the amount of samples that will be required, how frequently they will need to be tested, and how they should be handled and analyzed. A large sample size should be taken to capture the variability and trends in product quality over time. Still, the size should be moderate enough such that it is not impracticable or expensive. A sampling frequency is also required for the product's shelf life and quality changes. Proper sampling techniques are also required to be taken for guaranteeing the integrity and characteristics of the samples.

4. **Determining the methods for analysis :** The fourth stage is for analysis of the methods used for checking the quality of the products. These methods could be in the form of chemical, microbiological, physical, or certain sensory approaches which are based on the criteria and goals of the overall project. The methods used should be accurate, precise, and proven along with their conditions. It is also necessary to follow the appropriate protocols and procedures while testing and documenting the outcomes.

5. **Evaluation of information :** The fifth stage is analyzing the found data and interpreting the conclusions made from it. Appropriate statistical methods and instruments should be used for the condensing and modeling of the data. Various sources which may lead to uncertainty and variation in the data should also be identified, this includes environmental variations, measurement errors, and sampling errors. Certain patterns, trends, and correlations should be determined between the quality parameters and storage circumstances. The importance and application of the results regarding the standards and goals also need to be assessed.

6. **Reporting the final results :** The last stage is presenting the findings and discussing the implications of the overall analysis of the study. A proper and thorough report needs to be written which includes the goals, procedures, information, analysis, and findings of the research. In addition, the recommendations and implications used in the study also need to be included for product creation, improvement, and management. The report needs to be presented in a style that is appropriate for the target audience and which clearly determines the goals.

5.2 Coding Standards

1. **Coding Style:** The code should be written in a modular and reusable way to ensure code reusability and maintainability.

2. **Documentation:** The documentation of the code should be clear and concise, and comments should also be included within the code to explain certain statements and docstrings should be created for functions and classes.

3. **Use of Libraries:** Python libraries like TensorFlow, PyTorch, OpenCV, and scikit-learn should be used to leverage the existing functionalities and optimize the code's performance.

4. **Error Handling:** Implementation of error handlers should be done to handle exceptions and errors that may occur during the execution of the code.

5. **Testing:** Unit tests should be created to verify that the code is functional and for checking whether the predictive model created is operating as intended or not.

6. **Version Control:** Keeping track of changes, working together with the team members, and preserving the record of the project should be done using control systems, such as Git.

7. **Code Review:** The code should be reviewed to make sure it is well-written and follows the coding guidelines. Along with this, feedback should be taken after review for any changes to be done.

5.3 Testing Standards

1. **Experiment Tracking :** Experiment tracking is the process used for systematically organizing and managing machine learning experiments, which includes parameters, metrics, and results, using tools such as MLflow or Neptune. These technologies help to provide a single platform that records the experiment details, which makes it easier to reproduce, collaborate, and compare several model iterations. To increase model accuracy and generalization, the researchers can easily and efficiently track the model performance over time, spot the patterns, and make data-driven decisions by keeping a clear record of experiments.

2. **TensorBoard :** TensorBoard is offered by TensorFlow. It is a visualization tool used for examining and tracking deep learning models. It provides tools, which can be used for monitoring training progress, analyzing model performance, and displaying model graphs. Users are able to analyze the model behavior in an insightful manner by visualizing metrics like accuracy, loss curves, and other performance indicators. It makes it possible to visualize the data flow across the computational graph, making it easier to troubleshoot and optimize models.

3. **Summary Writer :** TensorBoard is used to show the appropriate information that is logged during model training in TensorFlow's Summary Writer tool. Various metrics, including gradients, activations, loss values, and accuracy, can be recorded by users to monitor the model's behavior and performance over time. Researchers can better understand the training process, spot possible problems, and make decisions for enhancing the model's performance by using the Summary Writer.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

In conclusion, this project has demonstrated the potential of machine learning and deep learning technologies in transforming the way the agricultural and food processing industries predict the shelf life of fruits and vegetables. By moving away from traditional, subjective methods and embracing data-driven, scientific approaches, we have shown that it is possible to increase the accuracy and efficiency of shelf life prediction.

The use of diverse datasets, rigorous preprocessing, and advanced modeling techniques has resulted in models that outperform traditional methods in key performance metrics such as accuracy, precision, and recall. These models not only improve inventory management and reduce waste but also contribute to increased profitability for producers and suppliers.

Furthermore, the implications of this project extend beyond the immediate benefits to the industry. By reducing food waste and optimizing supply chains, these predictive models also contribute to sustainability efforts and the broader goal of ensuring food security.

While the results of this project are promising, it is important to note that the field of shelf life prediction is continually evolving, and there is always room for improvement and innovation. As we continue to refine our models and incorporate new data and techniques, we look forward to contributing further to this vital area of research.

Overall, this project underscores the transformative potential of data science in addressing critical challenges in the agricultural and food processing industries, paving the way for a future where technology and data-driven decision-making becomes the norm rather than the exception.

6.2 Future scope

The future scope of this project is vast and promising. Here are some potential directions:

1. **Expansion of Dataset**: The current models are trained on a specific set of fruits and vegetables. In the future, we can expand the dataset to include a wider variety of produce, which would make the models more universally applicable.
2. **Integration of IoT Devices**: The project could benefit from the integration of Internet of Things (IoT) devices for real-time data collection. Sensors could be used to monitor and record changes in temperature, humidity, and other environmental factors that affect the shelf life of fruits and vegetables.
3. **Improvement of Models**: While the current models have shown promising results, there is always room for improvement. Future work could explore the use of more advanced machine learning and deep learning techniques to further enhance the accuracy of the shelf life prediction.
4. **Development of a User-Friendly Interface**: To make these models accessible to producers and suppliers, future work could involve the development of a user-friendly interface that allows users to easily input data and receive predictions.
5. **Collaboration with Industry Partners**: Collaborating with industry partners could help to validate and refine the models. This would also provide opportunities to pilot the models in real-world settings and gather feedback for further improvements.
6. **Sustainability and Food Security**: In the long term, this project could contribute to global sustainability efforts and food security. By reducing food waste and optimizing supply chains, we can make better use of our resources and ensure that fresh produce reaches consumers in a timely manner.

In conclusion, the future scope of this project is extensive and holds the potential to revolutionize the agricultural and food processing industries. As we continue to innovate and refine our models, we look forward to seeing the positive impact of our work.

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INDIVIDUAL CONTRIBUTION REPORT:

Predicting shelf life of fruits

Divita Topno
2105455

Abstract: The aim of this project is to predict the shelf life of fruits and vegetables using deep learning techniques. The project involves collecting and preprocessing data, applying deep learning models, and analyzing the results to determine the effectiveness of the approach.

Individual contribution and findings:

Prepared the collected data for testing and training. Used Python script to import the downloaded images into a singular folder to compress all the images; these compressed images were further saved in another folder for easier access. Also, used another Python script to split the final data into two datasets, which were used for training and testing. At last, loaded all the finalized data on GitHub and also imported them to Colab Notebook.

Individual contribution to project report preparation:

Contributed to Chapter 5: Standards Adopted; the Design Standards that were used for the actual project to be developed, the Coding Standards that were adopted for the coding process, and the Testing Standards used during the testing of the model using the collected data.

Individual contribution to project presentation and demonstration:

Contributed to the project presentation by explaining the process of how the downloaded images were compressed and divided into two datasets for training and testing purposes. Also the process of uploading the finalized data into GitHub and Colab Notebook.

Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT:

Predicting shelf life of fruits

Satvik Agrawal
21052869

Abstract: The aim of this project is to predict the shelf life of fruits and vegetables using deep learning techniques. The project involves collecting and preprocessing data, applying deep learning models, and analyzing the results to determine the effectiveness of the approach.

Individual contribution and findings:

Used Gradio library, a Python library used for building machine learning applications or for query testing; to deploy the model created using Python script and tested it by using various inputs to check the accuracy and functionality of the model.

Individual contribution to project report preparation:

Contributed to Chapter 3: Problem Statement/Requirement Specifications; the Project Planning that included the steps for model creation, project analysis, and system design, where design constraints were described and the block diagram of the code was made.

Individual contribution to project presentation and demonstration:

Contributed to the project presentation by deploying it in a website with a beautiful and interactive UI allowing users to upload images using their camera as well as from their local storage and get predictions with probabilities for each prediction.

Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT:

Predicting shelf life of fruits

Zoyah Afsheen Sayeed
21052990

Abstract: The aim of this project is to predict the shelf life of fruits and vegetables using deep learning techniques. The project involves collecting and preprocessing data, applying deep learning models, and analyzing the results to determine the effectiveness of the approach.

Individual contribution and findings:

Contributed to the data visualization and model evaluation. Specifically, utilized Python scripting to generate line graphs that depict the loss and accuracy curves of two Convolutional Neural Network (CNN) models: TinyVGG and EfficientNet-B0. These graphs provide a clear visual representation of each model's performance over time, allowing for an in-depth analysis of their learning process and effectiveness in classifying images into object categories. Furthermore, created a confusion matrix for the EfficientNet-B0 model which helped us understand our model's performance. It showed where the model did well and where it needs improvement. This was key in making our image classification system better and more accurate. Deployed the working Gradio model on HuggingFace.

Individual contribution to project report preparation:

Contributed to Chapter 4: Implementation, the methodology used, training and testing of the model, and result analysis of the entire model, and Chapter 6: Conclusion and Future Scope, along with proofreading of the entire report.

Individual contribution to project presentation and demonstration:

Presented on data visualization and in-depth result analysis. Explained key metrics and showcased graphical representation of results. Demonstrated techniques like confusion matrices and error visualizations to analyze model performance and identify patterns.

Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT:

Predicting shelf life of fruits

Md Ehteshamur Rahman
21053213

Abstract: The aim of this project is to predict the shelf life of fruits and vegetables using deep learning techniques. The project involves collecting and preprocessing data, applying deep learning models, and analyzing the results to determine the effectiveness of the approach.

Individual contribution and findings: Collected the data used in the project. Instead of using Kaggle datasets, created a custom dataset from Bing Images. This dataset included images of very brown bananas and overripe apples. Used a Python script to download 100 images per fruit category, resulting in a total of 1000 images for the project. This approach ensured that the dataset was tailored to the specific needs of the project. Gained valuable experience in data collection and preprocessing. Learned how to use Python libraries to download images and organize them into a suitable dataset for deep learning tasks. This hands-on experience will be beneficial for future projects involving data collection and preprocessing.

Individual contribution to project report preparation:
Contributed to Chapter 1: Introduction, giving an overview of the problem statement at hand and the solution provided, and Chapter 2: Basic Concepts/Literature Review of the project report by providing details about the machine learning, deep learning and data analysis techniques used.

Individual Contribution for Project Presentation and Demonstration:
Prepared slides to showcase the data collection process. Explained the importance of using a unique dataset for our project and highlighted the challenges and solutions we encountered during the data collection phase. Demonstrated how the dataset will be used in training and evaluating our deep learning models.

Full Signature of Supervisor: …….

Full signature of the student: ……..

INDIVIDUAL CONTRIBUTION REPORT:

Predicting shelf life of fruits

Abhinav Aakash
21053242

Abstract: The aim of this project is to predict the shelf life of fruits and vegetables using deep learning techniques. The project involves collecting and preprocessing data, applying deep learning models, and analyzing the results to determine the effectiveness of the approach.

Individual contribution and findings:

Investigated convolutional neural networks (CNNs) by implementing two models. A Tiny VGG model was constructed to analyze the influence of network depth on performance and its ability to capture spatial information in images. This exploration provided insights into fundamental CNN concepts. Subsequently, transfer learning was leveraged to implement the complex EfficientNet-B0 architecture, a powerful model widely used for image classification tasks. This accomplishment demonstrates expertise in both core CNN principles and the application of advanced techniques, showcasing the ability to build and utilize sophisticated models for real-world computer vision tasks.

Individual contribution to project report preparation:

Contributed towards writing the Abstract along with formatting and Chapter 6: Conclusion and Future Scope, along with proofreading of the entire report.

Individual contribution to project presentation and demonstration:

Prepared slides explaining various deep learning architectures and the rationale behind choosing the specific one for our project. Demonstrated the training pipeline, showcasing the selected architecture and its strengths for the task at hand. Highlighted how architecture selection plays a crucial role in achieving optimal model performance.

Full Signature of Supervisor: …….

Full signature of the student: ……..

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Predicting shelf life of fruits

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